

Symbols

α_{s+}	The critical angle of attack at which stall occurs for positive angle of attack.
α_{s-}	The critical angle of attack at which stall occurs for negative angle of attack.
α	The angle of attack.
β	The sideslip angle.
λ	Prescribed trim parameters: airspeed and symmetric engine tilt. This symbol replaces the earlier parameter vector p to avoid confusion with roll rate..
$\mathbf{a}_{\text{trim}}$	Equilibrium actuator-command reference constructed from the selected trim curve..
$\mathbf{A}$	Jacobian of the reduced dynamics with respect to the reduced state at a trim..
$\mathbf{a}$	Actuator vector used by the selected problem: physical settings (propeller speed, elevator deflection and tilt) for geometric trim-path costs, or actuator commands for dynamics. Its interpretation is stated locally..
$\mathbf{B}$	Jacobian of the reduced dynamics with respect to actuator commands at a trim..
$\mathbf{F}_{\text{trim}}$	Three-component vector of horizontal-force, vertical-force and pitch-moment residuals. Its zero set defines the longitudinal trim family..
$\mathbf{f}$	Nonlinear state-derivative function of the aircraft and actuator model..
$\mathbf{J}_{\lambda}$	Jacobian of the trim residual with respect to the prescribed airspeed-tilt parameter vector..
$\mathbf{J}_w$	Jacobian of the trim residual with respect to the fixed-airspeed continuation variables..
$\mathbf{J}_y$	Jacobian of the trim residual with respect to the dependent variables y . Its nonsingularity permits local airspeed-tilt coordinates..
$\mathbf{Q}$	Positive-semidefinite state-deviation weighting matrix in trajectory optimization, with units chosen consistently with the state scaling..
$\mathbf{R}$	Positive-semidefinite actuator-command-deviation weighting matrix in trajectory optimization, with units chosen consistently with the input scaling..
$\mathbf{t}_c$	Unit tangent to a fixed-airspeed trim branch, evaluated in consistently scaled continuation coordinates..
$\mathbf{w}_c$	Variables continued at fixed airspeed: symmetric tilt, pitch, total thrust and elevator deflection. Distinct from body-axis vertical velocity w ..
$\mathbf{x}_r$	State vector used for local stability analysis after removing cyclic position coordinates and any other coordinates excluded from the selected stabilization task..
$\mathbf{y}$	Dependent trim variables: pitch attitude, total thrust and elevator deflection..
$\mathbf{z}$	Complete reduced trim-coordinate vector: airspeed, symmetric engine tilt, pitch attitude, total thrust and elevator deflection; distinct from global vertical position..
χ	Thrust direction above the horizontal flight path in a level trim, equal to symmetric engine tilt plus pitch attitude. Unit: radians..
$\delta\alpha$	The size of region where normal aerodynamics transition to post-stall aerodynamics.
$\Delta\ell$	Predictor step in scaled pseudo-arclength coordinates; distinct from the transition progress variable..

$\delta\eta_{E,\text{cmd}}$	Common commanded tilt of both nacelles, including actuator-lag feedforward during a transition. Unit: radians..
$\delta\eta E_l$	The commanded tilt of the left engine.
$\delta\eta E_r$	The commanded tilt of the right engine.
δE_{max}	Maximum permitted magnitude of elevator deflection for symmetric travel limits. Unit: radians..
δA	The deflection of the ailerons.
δE	The deflection of the elevator.
δN_l	The rev/s of the left propeller.
δN_r	The rev/s of the right propeller.
δR	The deflection of the rudder.
ϵ_J	Small positive dimensionless regularization constant in the reserve penalty; distinct from the nacelle-tilt departure..
$\eta_{E,\text{max}}$	Maximum permitted symmetric nacelle tilt. Unit: radians..
$\eta_{E,\text{min}}$	Minimum permitted symmetric nacelle tilt. Unit: radians..
ηE_l	The tilt of the left engine.
ηE_r	The tilt of the right engine.
ηE	The symmetric tilt of both engines.
Γ	Continuous curve in the complete reduced trim-coordinate space, connecting hover with a cruise trim..
$\mathcal{C}$	Set of feasible forward-flight endpoint trims with zero symmetric nacelle tilt and airspeed in the selected cruise interval..
$\mathcal{H}$	Family of exact hover trims parameterized by the elevator setting, which is aerodynamically ineffective at zero airspeed in this model..
$\mathcal{M}_{\text{feas}}$	Subset of the trim solution set satisfying the selected actuator, attitude, aerodynamic-validity and operational constraints..
$\mathcal{M}$	Reduced steady-trim solution set; locally a two-dimensional manifold where the full residual Jacobian has rank three..
$\mathcal{P}_{\text{shaft}}$	Total mechanical shaft power required by both propellers. Unit: watts..
RPM_{max}	Maximum permitted speed of each propeller, measured in revolutions per minute..
RPM	Common propeller speed measured in revolutions per minute; numerically equal to 60 times the speed in revolutions per second..
μ_E	Dimensionless fractional elevator-travel reserve, equal to zero at either symmetric elevator limit..
μ_T	Dimensionless fractional total-thrust reserve, equal to zero at the thrust ceiling..
μ	Minimum of the thrust and elevator reserve measures used for path selection; this is not a complete measure of control authority..
ω_{emax}	The maximum angular velocity of the servo+nacelle system.
$\bar{c}$	Mean aerodynamic chord.
ϕ	The roll of the craft on body axis x .
Π_{ij}	Penalty assigned to a graph connection for low actuator reserve along the connection..
ψ	The yaw of the craft on body axis z .
ρ	The density of air that the craft is in.
τ_ϕ	The moment acting around the roll axis.

τ_ψ	The moment acting around the yaw axis.
τ_θ	The moment acting around the pitch axis.
τ_{Aero}	The moment acting on the craft due to aerodynamic forces.
τ_{Diff}	The moment acting on the craft due to differential rotation speed of propellers.
τ_{Thrust}	The moment acting on the craft due to thrust forces.
τ_e	The time constant representing the servo+nacelle response.
$\theta_{\max}$	Maximum permitted pitch attitude in the selected operating domain. Unit: radians..
$\theta_{\min}$	Minimum permitted pitch attitude in the selected operating domain. Unit: radians..
θ	The pitch of the craft on body axis y .
ε	Departure of the symmetric nacelle tilt from vertical, defined as $\pi/2$ minus tilt. Unit: radians; distinct from the path-penalty regularization constant..
$\hat{\mathbf{F}}_{\text{trim}}$	Dimensionless trim residual: forces divided by weight and pitch moment divided by weight times mean aerodynamic chord..
ξ_{trim}	State reference constructed from the selected trim curve. For a full-state cost, cyclic positions must be supplied by a separate kinematic reference or assigned zero cost weight..
ξ	The state vector of the craft.
b	Wingspan.
$C_{D,\max}$	The maximum value of the drag coefficient.
$C_{d,\text{pre-stall}}(\alpha)$	The pre-stall drag coefficient.
c_{ij}	Nonnegative graph-search cost of a validated connection from trim vertex i to trim vertex j ..
$C_{l,\max}$	The maximum value of the lift coefficient.
$C_{l,\text{pre-stall}}(\alpha)$	The pre-stall lift coefficient.
C_{l_β}	The dihedral effect.
$C_{l_{\delta_A}}$	Aileron effectiveness.
$C_{l_{\delta_R}}$	Rudder-induced roll.
C_{l_r}	Yaw rate induced roll.
C_{l_p}	Roll damping.
C_{l_0}	Roll moment at trim.
C_{m_α}	Static longitudinal stability.
$C_{m_{\delta_E}}$	Elevator effectiveness.
C_{m_0}	Pitch moment at trim.
C_{m_q}	Pitch damping.
C_{n_β}	Weathercock stability.
$C_{n_{\delta_A}}$	Adverse yaw.
$C_{n_{\delta_R}}$	Rudder effectiveness.
C_{n_o}	Yaw moment at trim.
C_{n_p}	Roll-rate-induced yaw.
C_{n_r}	Yaw damping.
C_T	The thrust coefficient of the craft's propeller.
$C_d(\alpha)$	The coefficient of drag.
$C_l(\alpha)$	The coefficient of lift.
$C_y(\beta)$	The side-sweep coefficient.

D_{prop}	The diameter of the craft's propeller.
D	The force of drag acting on the craft along the x axis.
E_{Height}	The distance between the engines and CoM in the z axis..
E_{Span}	The distance between the engines and CoM in the y axis..
E_y	The distance between the CoM and the engine in y axis.
E_z	The distance between the CoM and the engine rotation axis in z axis.
F_{Aero}	The aerodynamical forces acting on the craft, includes F_{Aero_x} , F_{Aero_y} , F_{Aero_z} to divide into axes.
F_{Thrust}	The thrust forces acting on the craft, includes F_{Thrust_x} , F_{Thrust_y} , F_{Thrust_z} to divide into axes.
F_{Wind}	The aerodynamical forces acting on the craft, includes within the wind frame.
g_b	The gravitational forces acting on the craft.
G	Local map from airspeed and elevator deflection to pitch, total thrust and nacelle tilt near hover..
g	Magnitude of gravitational acceleration. Unit: metres per second squared. The existing entry F_{grav} denotes the body-frame vector, not this scalar..
H	Scalar force-equilibrium residual obtained by projecting the required thrust perpendicular to the thrust direction. Unit: newtons..
I_C	Accepted interval of cruise airspeeds used when the final airspeed is not prescribed exactly..
k_H	Coefficient of airspeed squared in the near-hover nacelle-tilt departure. Unit: radians times seconds squared per metre squared. This replaces the earlier k to distinguish it from an iteration index..
K_q	The torque coefficient of the craft's propeller.
L	The force of lift acting on the craft.
m_{craft}	The mass of the craft.
N	Common speed of both propellers in a symmetric trim, measured in revolutions per second..
P_{max}	Maximum total symmetric thrust available under the assumed constant thrust-coefficient model. Unit: newtons..
P	Total thrust of the two engines; distinct from shaft power. Unit: newtons..
p	The angular velocity of the craft along body axis x .
q_∞	Dynamic pressure.
Q_{prop}	Magnitude of the resisting shaft torque of one propeller. Unit: newton metres; distinct from the state-cost weighting matrix..
q	The angular velocity of the craft along body axis y .
r	The angular velocity of the craft along body axis z .
S	Wing reference area.
S	The surface area of the wing.
s	Dimensionless transition progress parameter ranging from zero at the start to one at the end..
T_f	Duration of the dynamically feasible transition. Unit: seconds..
u	The velocity in axis x of the craft within its body coordinates.
V_a	Airspeed.
v	The velocity in axis y of the craft within its body coordinates.
w_ℓ	Nonnegative weight of normalized edge length in the graph-search objective..

w_1	Nonnegative weight of the squared first derivative of the normalized trim path..
w_2	Nonnegative weight of the squared second derivative of the normalized trim path..
w_3	Nonnegative weight of the actuator-reserve penalty in the continuous path objective..
w_r	Nonnegative weight of the low-reserve penalty in the graph-search objective..
w_u	Nonnegative weight of actuator-setting variation in the graph-search objective..
w	The velocity in axis z of the craft within its body coordinates.
x_E	The x position of the craft in the global coordinates.
X	The sum of all forces acting along axis x .
y_E	The y position of the craft in the global coordinates.
Y	The force of drag acting on the craft along the y axis.
Y	The sum of all forces acting along axis y .
z_E	The z position of the craft in the global coordinates.
Z	The sum of all forces acting along axis z .

Chapter 1

Dynamics

1.1 Overview

When considering a standard UAV dynamics the plant can be simplified to a 6 DoF rigid body where forces act on a point moving within space. In case of 2 engine tilt rotor design the system expands to a 6 + 2 DoF, where the two additional degrees of freedom represent the tilt of left and right engine nacelle. The entire system can still be simplified with following assumptions:

- The plant is a ideal rigid body, meaning the body maintains constant geometry. The angling of the nacelles and deflection of the control surfaces will be only considered in form of forces and torques acting on the body.
- The torque acting in the Y-axis generated by rotation of engine nacelles is negligible.
- The rotational complexity is simplified into a 3 x 3 mass moment of inertia tensor rotating around the center of mass.
- The mass of the plant remains constant.
- The body of the aircraft is symmetric in X - Z Plane, simplifying the inertia matrix.

The dynamics equations will be derived with Newton - Euler method as it allows an easy, step by step process to account for forces acting in the UAV. For a 6-DoF system the translational and rotational equation of motion within Newton - Euler method are as follow:

- Translational motion - the linear accelerations of the center of mass of the plant follow $F_b = m(\dot{V}_b + \omega_b \times V_b)$, where F_b are the forces acting on the body, V_b are the velocities within body frame and ω_b are the angular velocities. The linear accelerations are derived from the Newton's Law.
- Rotational motion - the rotation of the body that accounts for gyroscopic coupling through $M_b = I\dot{\omega}_b + \omega_b \times (I\omega_b)$, where M_b are the torques acting on the plant and I that is the 3 x 3 inertia matrix.

The dynamic of the plant will follow the NED coordinate convention with Z-axis pointing down and following the NEDs right handed rule and the GP-LV (Global Position - Local Velocity) convention where velocities are calculated within the body frame of the plant and positions exists within the global frame where point (0, 0, 0) is defined as position of the plant at $t = 0$.

The state-vector of the plant will be described as:

$$\xi = \begin{bmatrix} R_1^T & R_2^T & v_1^T & v_2^T & E_t^T \end{bmatrix}^T = \begin{bmatrix} x_E & y_E & z_E & \phi & \theta & \psi & u & v & w & p & q & r & \eta E_r & \eta E_l \end{bmatrix}^T \quad (1.1)$$

Where:

- $R_1^T = \begin{bmatrix} x_E & y_E & z_E \end{bmatrix}^T$ is the global position vector of the craft
- $R_2^T = \begin{bmatrix} \phi & \theta & \psi \end{bmatrix}^T$ is the vector containing the three euler angles (roll, pitch, yaw) in the inertial frame
- $v_1^T = \begin{bmatrix} u & v & w \end{bmatrix}^T$ is the velocity vector within the body frame of the craft
- $v_2^T = \begin{bmatrix} p & q & r \end{bmatrix}^T$ is the vector containing the angular velocities.
- $E_t^T = \begin{bmatrix} \eta E_r & \eta E_l \end{bmatrix}^T$ is the vector containing the throttle of the engines.

The orientation of each are showcased in 1.1, 1.2 and 1.3.

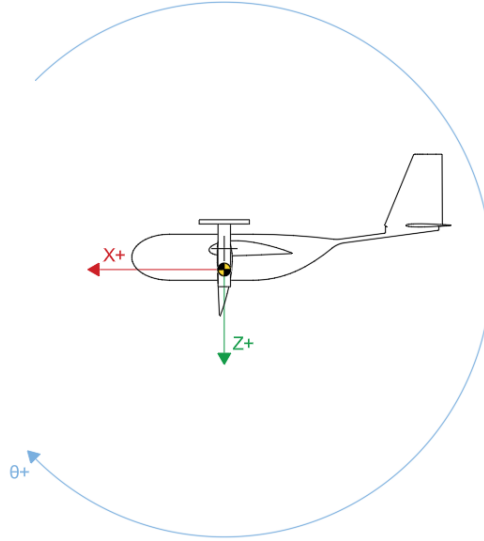


FIGURE 1.1: The axes within the body frame as visible from the side.

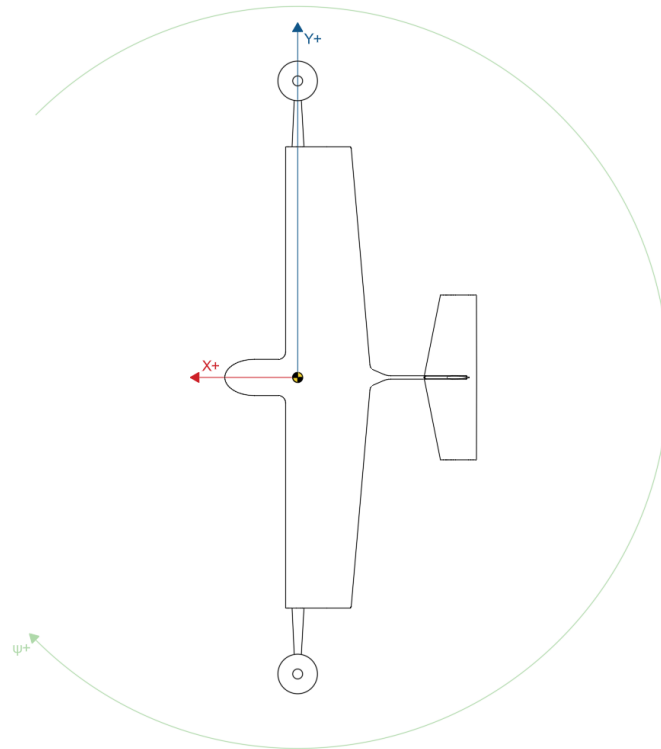


FIGURE 1.2: The axes within the body frame as visible from the top.

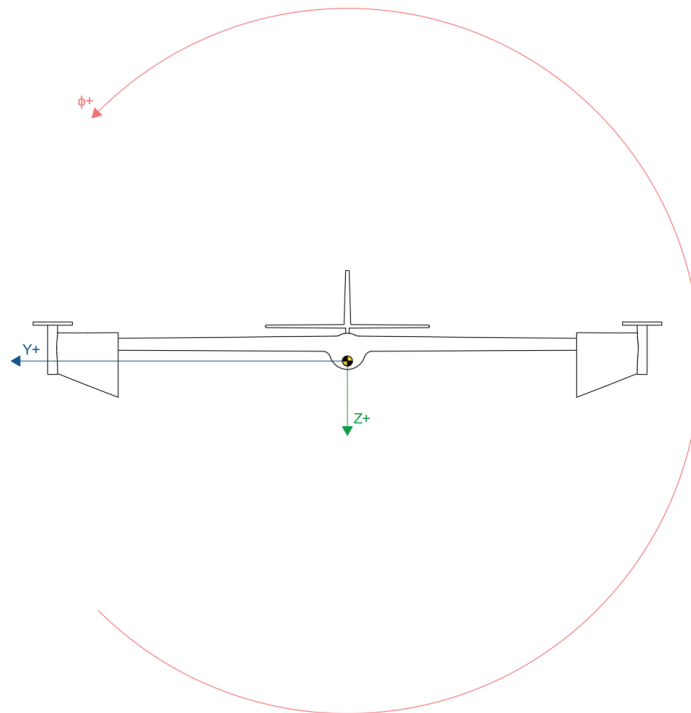


FIGURE 1.3: The axes within the body frame as visible from the front.

1.2 The delayed response of the engine tilting mechanism.

While the small size of the systems this model is intended to be used for means that response from the tilt mechanism is near instantaneous it also means that response of the airframe to any disturbance is also extremely fast and the fast actuator dynamics become unlikely to achieve the 10:1 time scale separation ratio in relation to slow plant dynamics where the actuators dynamics can be safely omitted. To model the delay of the tilting mechanism a rate-limited first-order model will be used in form:

$$\dot{\theta} = clip(\frac{1}{\tau}(\theta_{cmd} - \theta), -\omega_{max}, +\omega_{max}) \quad (1.2)$$

This results in two equations:

$$\eta \dot{E}_r = clip(\frac{1}{\tau_e}(\delta \eta E_r - \eta E_r), -\omega_{emax}, +\omega_{emax}) \quad (1.3)$$

$$\eta \dot{E}_l = clip(\frac{1}{\tau_e}(\delta \eta E_l - \eta E_l), -\omega_{emax}, +\omega_{emax}) \quad (1.4)$$

1.3 Forces acting on the model

When studying the movement of rigid bodies within the framework of the Newton - Euler method the accelerations of the object are obtained from the forces acting on the craft via Newtons 2nd law. But because the craft will be operating within a rotating, body-fixed frame the kinematic transport theorem needs to be used resulting in Newtons 2nd law being expressed as $\vec{F} = m(\frac{d\vec{v}}{dt}) + m(\vec{\omega} \times \vec{v})$. When solving this equation for $\frac{d\vec{v}}{dt}$ the forces translate to acceleration in following way:

$$\begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{bmatrix} = \begin{bmatrix} rv - qw + \frac{X}{m_{craft}} \\ pw - ru + \frac{Y}{m_{craft}} \\ qu - pv + \frac{Z}{m_{craft}} \end{bmatrix} \quad (1.5)$$

The sum of forces acting on the craft consists of the aerodynamic forces, thrust forces and gravity:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} F_{Thrustx} + F_{Aerox} \\ F_{Thrusty} + F_{Aero y} \\ F_{Thrustz} + F_{Aero z} \end{bmatrix} + m_{craft} g_b \quad (1.6)$$

1.4 Force of gravity

The force of gravity within the inertial earth frame takes form $g_E = \begin{bmatrix} 0 & 0 & g \end{bmatrix}^T$. To apply it within the body frame of the craft it needs to be multiplied by a rotation matrix resulting in form:

$$g_b = g \begin{bmatrix} -\sin\theta \\ \sin\phi\cos\theta \\ \cos\phi\cos\theta \end{bmatrix} \quad (1.7)$$

1.5 Force of thrust

The force of thrust generated by the propellers can be described in relation to the rotation speed of the propellers as:

$$T(N) = C_T \rho N^2 D_{prop}^4 \quad (1.8)$$

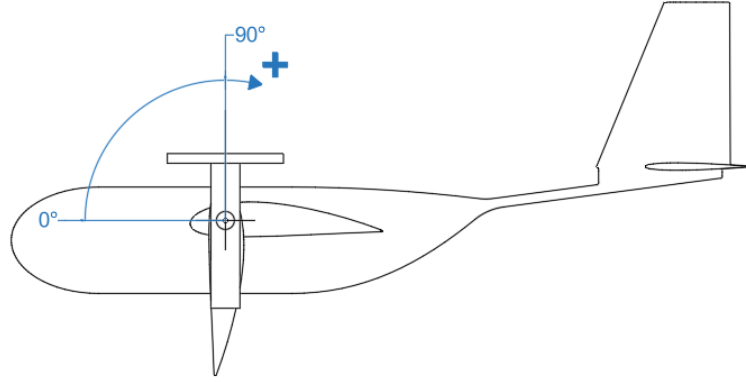


FIGURE 1.4: The angle of engine tilt.

The D_{prop} and C_T are parameters of the crafts construction. The air density ρ can also be treated as constant due to legal limits on drone flight altitude. During high speed flight the efficiency of propellers drops but due to limited speeds of drones this will also be simplified.

The forces acting on the craft via engines can be described as:

$$F_{Thrustx} = T(\delta N_r) \cos \eta E_r + T(\delta N_l) \cos \eta E_l \quad (1.9)$$

$$F_{Thrusty} = 0 \quad (1.10)$$

$$F_{Thrustz} = -T(\delta N_r) \sin \eta E_r - T(\delta N_l) \sin \eta E_l \quad (1.11)$$

The forces acting in each axis depend on tilting of each engine. As each engine needs to be able to tilt independently to allow for control over the craft the force of each needs to be calculated independently and summed. The force in axis z achieves maximum at 90° tilt and zero at 0° tilt. The force in axis x behaves inversly. At no engine position the force acts in y axis. As the z axis points downwards the forces acting on it must be inverted.

1.6 Aerodynamic Forces

Aerodynamic forces act withing the airflow frame that is rotated by α and β in relation to the body frame. Within this frame the forces acting upon the plant are:

- Lift L - Acts perpendicular to the relative airspeed in the symmetry plane, countering gravity.
- Drag D - Acts parallel and opposite to the relative airspeed, resisting forward motion.
- Sideforce Y - Acts along the lateral axis introducing sideslips and crosswinds.

The aerodynamic forces act in relation to the True Air Speed that can be defined as the magnitude of the relative velocity vector between the aircraft center of mass and the surrounding mass of air. This can be acquired from equation:

$$V_a = \sqrt{u^2 + v^2 + w^2} \quad (1.12)$$

The angle α is the angle between longitudinal body axis x and the relative wind velocity vector projected onto planes x - z symmetry plane. The angle α affects lift and longitudinal stability of the craft. It can be acquired from relation of the aircraft w and u velocities.

$$\alpha = \arctan2(w, u) \quad (1.13)$$

The angle β is defined as the angle between the total relative wind velocity vector and the aircraft's plane of symmetry (x - z plane) and can be acquired from:

$$\beta = \text{atan2}(v, \sqrt{w^2 + u^2}) \quad (1.14)$$

It must be noted that when airspeed becomes 0 at hover the α and β becomes physically meaningless and behavior of these equations is undefined. When implementing this model a convention for zero air speed should be implemented. For a VTOL craft accurate modeling of post stall behavior at extreme α angles becomes important and thus must be modeled. To model the aerodynamic forces in post stall conditions a modified Viterna-Corrigan model will be used, where C_l becomes $\frac{1}{2}C_{d,max}\sin(2\alpha)$ and C_d becomes $C_{D,max}\sin^2\alpha$. The models will be blended with standard, table derived C_l and C_d coefficients via function:

$$\sigma = \frac{1}{2}(2 + \tanh(\frac{\alpha - \alpha_{s+}}{\delta\alpha}) + \tanh(\frac{\alpha - \alpha_{s-}}{-\delta\alpha})) \quad (1.15)$$

The lift and drag coefficients thus become:

$$C_l(\alpha) = (1 - \sigma)(C_{l,pre-stall}(\alpha)) + \sigma(\frac{1}{2}C_{D,max}\sin(2\alpha)) \quad (1.16)$$

$$C_d(\alpha) = (1 - \sigma)(C_{d,pre-stall}(\alpha)) + \sigma(C_{D,max}\sin^2\alpha) \quad (1.17)$$

The aerodynamic within the wind frame can be defined as:

$$L = \frac{1}{2}\rho V_a^2 S C_l(\alpha) \quad (1.18)$$

$$D = \frac{1}{2}\rho V_a^2 S C_d(\alpha) \quad (1.19)$$

$$Y = \frac{1}{2}\rho V_a^2 S C_y(\beta) \quad (1.20)$$

These can be organized within a vector F_{Wind} . Due to force D acting opposing to the velocity u and force L acting opposing to the gravity these have to be inverted.

$$F_{Wind} = \begin{bmatrix} -D \\ Y \\ -L \end{bmatrix} \quad (1.21)$$

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To apply these forces to the body frame the forces have to be rotated by α and β :

$$F_{Aero} = R_y(\alpha)R_z(\beta)F_{Wind} \quad (1.22)$$

Where the transformation matrix is:

$$R_y(\alpha)R_z(\beta) = \begin{bmatrix} \cos\alpha\cos\beta & -\cos\alpha\sin\beta & -\sin\alpha \\ \sin\beta & \cos\beta & 0 \\ \sin\alpha\cos\beta & -\sin\alpha\sin\beta & \cos\alpha \end{bmatrix} \quad (1.23)$$

Applying these rotation results in component forces:

$$F_{Aerox} = -D\cos\alpha\cos\beta - Y\cos\alpha\sin\beta + L\sin\alpha \quad (1.24)$$

$$F_{Aero y} = -D\sin\beta + Y\cos\beta \quad (1.25)$$

$$F_{Aero z} = -D\sin\alpha\cos\beta - Y\sin\alpha\sin\beta - L\cos\alpha \quad (1.26)$$

1.7 Torques acting on the model

Within the Newton Euler method the torques are included via the Euler's Rotational Equation.

$$\tau = I\dot{\omega} + \omega \times (I\omega) \quad (1.27)$$

The two distinct parts of the equation account for the torque required to produce angular acceleration ($I\dot{\omega}$) and changing direction of the angular momentum vector due to rotation of the body frame in form of the cross product ($\omega \times (I\omega)$).

To use it within a model the vector cross product must be expanded with the inertial matrix of the aircraft and the resulting equation must be solved for $\dot{\omega}_x$, $\dot{\omega}_y$ and $\dot{\omega}_z$

$$I = \begin{bmatrix} I_{xx} & 0 & -I_{xz} \\ 0 & I_{yy} & 0 \\ -I_{zx} & 0 & I_{zz} \end{bmatrix} \quad (1.28)$$

This inertia matrix is the result of aircraft's symmetry in the x - z plane. This reduces products of inertia I_{xy} and I_{yz} to zero but leaves a dynamic coupling between roll and yaw through their accelerations. When this matrix expands the equation it becomes:

$$\tau_x = I_{xx}\dot{\omega}_x - I_{xz}\dot{\omega}_z + (I_{zz} - I_{yy})\omega_y\omega_z - I_{xz}\omega_x\omega_y \quad (1.29)$$

$$\tau_y = I_{yy}\dot{\omega}_y + (I_{xx} - I_{zz})\omega_x\omega_z + I_{xz}(\omega_x^2 - \omega_z^2) \quad (1.30)$$

$$\tau_z = -I_{xz}\dot{\omega}_x + I_{zz}\dot{\omega}_z + (I_{yy} - I_{xx})\omega_x\omega_y + I_{xz}\omega_y\omega_z \quad (1.31)$$

Computation of $\dot{\omega}_y$ is trivial due to simple coupling with other axes resulting in:

$$\dot{\omega}_y = \frac{\tau_y + (I_{zz} - I_{xx})\omega_x\omega_z - I_{xz}(\omega_x^2 - \omega_z^2)}{I_{yy}} \quad (1.32)$$

The equations for x and z axis share both $\dot{\omega}_x$ and $\dot{\omega}_z$ resulting in equations in form:

$$\dot{\omega}_x = \frac{I_{zz}L' + I_{xz}N'}{I_{xx}I_{zz} - I_{xz}^2} \quad (1.33)$$

$$\dot{\omega}_z = \frac{I_{xz}L' + I_{xx}N'}{I_{xx}I_{zz} - I_{xz}^2} \quad (1.34)$$

Where:

$$\begin{bmatrix} L' \\ N' \end{bmatrix} = \begin{bmatrix} I_{xx} & -I_{xz} \\ -I_{xz} & I_{zz} \end{bmatrix} \begin{bmatrix} \dot{\omega}_x \\ \dot{\omega}_z \end{bmatrix} \quad (1.35)$$

Resulting in:

$$L' = \tau_\phi - (I_{zz} - I_{yy})qr + I_{xz}pq \quad (1.36)$$

$$N' = \tau_\psi - (I_{yy} - I_{xx})pq - I_{xz}qr \quad (1.37)$$

These equations can be ordered into a simplified form where most inertial terms are packed into coefficients:

$$\begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} c_1pq + c_2rq + c_7\tau_\phi + c_9\tau_\psi \\ c_3(p^2 - r^2) + c_4pr + c_8\tau_\theta \\ c_5pq + c_6rq + c_9\tau_\phi + c_{10}\tau_\psi \end{bmatrix} \quad (1.38)$$

Where:

$$c_1 = \frac{-I_{zz}I_{xz} - I_{xz}(I_{xx} - I_{yy})}{I_{xz}^2 - I_{xx}I_{zz}}$$

$$c_2 = \frac{-I_{zz}(I_{yy} - I_{zz}) + I_{xz}^2}{I_{xz}^2 - I_{xx}I_{zz}}$$

$$c_3 = -\frac{I_{xz}}{I_{yy}}$$

$$c_4 = -\frac{I_{xx} - I_{zz}}{I_{yy}}$$

$$c_5 = \frac{-I_{xz}^2 - I_{xx}(I_{xx} - I_{yy})}{I_{xz}^2 - I_{xx}I_{zz}}$$

$$c_6 = \frac{I_{xx}I_{xz} - I_{xz}(I_{yy} - I_{zz})}{I_{xz}^2 - I_{xx}I_{zz}}$$

$$c_7 = -\frac{I_{zz}}{I_{xz}^2 - I_{xx}I_{zz}}$$

$$c_8 = \frac{1}{I_{yy}}$$

$$c_9 = -\frac{I_{xz}}{I_{xz}^2 - I_{xx}I_{zz}}$$

$$c_{10} = -\frac{I_{xx}}{I_{xz}^2 - I_{xx}I_{zz}}$$

The sum of torques acting on the craft consists of the aerodynamic torques, torques resulting from thrust and torques as result of differential angular velocities of the spinning propellers:

$$\begin{bmatrix} \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} = \begin{bmatrix} \tau_{Aerox} + \tau_{Thrustx} + \tau_{Diffx} \\ \tau_{Aero y} + \tau_{Thrusty} + \tau_{Dif fy} \\ \tau_{Aero z} + \tau_{Thrustz} + \tau_{Diffz} \end{bmatrix} \quad (1.39)$$

1.8 Torques resulting from thrust

In hover or very-low-airspeed operation, the freestream aerodynamic forces scale toward zero with dynamic pressure, and become numerically ill-conditioned. Conventional fixed-wing stability-derivative models therefore cease to be a useful description of the aerodynamic forces. In the assumed craft geometry control surfaces are not immersed in rotor slipstream and thus provide near zero amount of control. In pure hover the craft relies instead on it's engines both to stay in the air and for control.

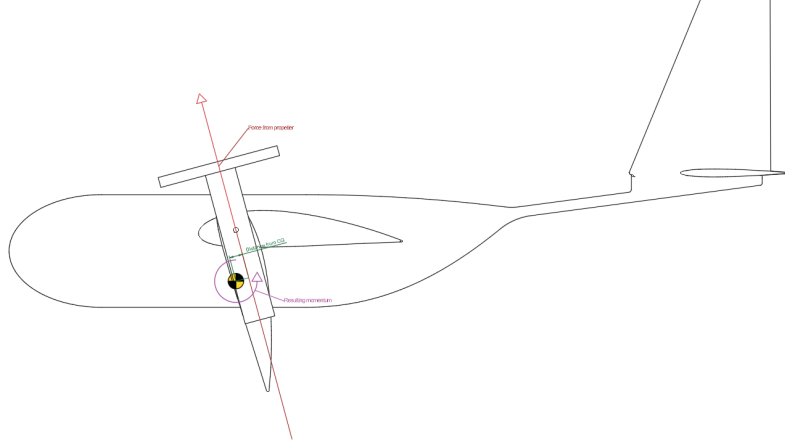


FIGURE 1.5: The side view of thrust creating a pitching torque.

The thrusts of engines are offset in z and y axes creating torques with their rotation. The component forces 1.9, 1.10 and 1.11 generated by the engines can be arranged into a vector and multiplied by vector of distances from CoM arranged into a skew-symmetric cross-product matrix:

$$\vec{\tau} = [\vec{r}] \times \vec{F} = \begin{bmatrix} 0 & -r_z & r_y \\ r_z & 0 & -r_x \\ -r_y & r_x & 0 \end{bmatrix} \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} \quad (1.40)$$

This can also be arranged into following form to derive component torques:

$$\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix} \quad (1.41)$$

Due to the studied model being powered by two engines these must be expanded into:

$$\vec{\tau} = \vec{r}_r \times \vec{F}_r + \vec{r}_l \times \vec{F}_l = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r_{rx} & r_{ry} & r_{rz} \\ F_{rx} & F_{ry} & F_{rz} \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r_{lx} & r_{ly} & r_{lz} \\ F_{lx} & F_{ly} & F_{lz} \end{vmatrix} \quad (1.42)$$

The final form for the component forces is:

$$\vec{\tau} = \vec{r}_r \times \vec{F}_r + \vec{r}_l \times \vec{F}_l = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r_{rx} & r_{ry} & r_{rz} \\ F_{rx} & F_{ry} & F_{rz} \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r_{lx} & r_{ly} & r_{lz} \\ F_{lx} & F_{ly} & F_{lz} \end{vmatrix} \quad (1.43)$$

$$\tau_{Thrustx} = (E_{Span} * -(T(\delta N_r, \rho) \sin(\eta E_r))) + (E_{Span} * (T(\delta N_l, \rho) \sin(\eta E_l))) \quad (1.44)$$

$$\tau_{Thrusty} = (E_{Height} * (T(\delta N_r, \rho) \cos(\eta E_r))) + (E_{Height} * (T(\delta N_l, \rho) \cos(\eta E_l))) \quad (1.45)$$

$$\tau_{Thrustz} = -(E_{Span} * (T(\delta N_r, \rho) \cos(\eta E_r))) - (-E_{Span} * (T(\delta N_l, \rho) \cos(\eta E_l))) \quad (1.46)$$

For the assumed geometry in hover, the vertical offset between the thrust-vector rotation axis and the center of mass provides an important pitch-control moment when the thrust vector develops a longitudinal component. This offset results in rotation of the engine generating a torque as

showcased in 1.5. Without this offset the craft of the assumed configuration would be uncontrollable in pitch.

Besides of torque created by differential engine tilt the torque in the y axis is also affected by torque created by spinning propellers. This torque is described by equation:

$$Q = \rho * n^2 * D^5 * K_q \quad (1.47)$$

The model assumes counter-rotating propellers where the torque produced by both propellers will be opposed and will result in:

$$\tau_{Diffx} = (\rho * \delta N_r^2 * D_{prop}^5 * K_q) \cos \eta E_r - (\rho * \delta N_l^2 * D_{prop}^5 * K_q) \cos \eta E_l \quad (1.48)$$

$$\tau_{Diffy} = 0 \quad (1.49)$$

$$\tau_{Diffz} = (\rho * \delta N_r^2 * D_{prop}^5 * K_q) \sin \eta E_r - (\rho * \delta N_l^2 * D_{prop}^5 * K_q) \sin \eta E_l \quad (1.50)$$

1.9 Torques resulting from aerodynamic forces

The torques acting on the craft in result of the aerodynamic forces in forms of control surface deflections and other aerodynamic effects can be described in standard stability analysis form:

$$\tau_{Aerox} = q_\infty S b (C_{l0} + C_{l\beta} \beta + C_{l\bar{p}} \bar{p} + C_{l\bar{r}} \bar{r} + C_{l\delta_A} \delta A + C_{l\delta_R} \delta R) \quad (1.51)$$

$$\tau_{Aero y} = q_\infty S \bar{c} (C_{m0} + C_{m\alpha} \alpha + C_{m\bar{q}} \bar{q} + C_{m\delta_E} \delta E) \quad (1.52)$$

$$\tau_{Aero z} = q_\infty S b (C_{n0} + C_{n\beta} \beta + C_{n\bar{p}} \bar{p} + C_{n\bar{r}} \bar{r} + C_{n\delta_A} \delta A + C_{n\delta_R} \delta R) \quad (1.53)$$

Where q_∞ is the dynamic pressure described as:

$$q_\infty = \frac{1}{2} \rho V_a^2 \quad (1.54)$$

And the non-dimensional angular rates are:

$$\bar{p} = \frac{pb}{2V_a} \quad (1.55)$$

$$\bar{q} = \frac{q\bar{c}}{2V_a} \quad (1.56)$$

$$\bar{r} = \frac{rb}{2V_a} \quad (1.57)$$

This form however results in singularities at $V_a = 0$ making the classical stability derivative models unusable in hover. The problematic V_a contained in the non-dimensional angular rates can be algebraically canceled with V_a appearing in the dynamic pressure equation resulting in following equations:

$$\tau_{Aerox} = \frac{1}{2} \rho V_a^2 S b (C_{l0} + C_{l\beta} \beta + C_{l\delta_R} \delta R + C_{l\delta_A} \delta A) + \frac{1}{4} \rho V_a S b^2 (C_{l_p} p + C_{l_r} r) \quad (1.58)$$

$$\tau_{Aero y} = \frac{1}{2} \rho V_a^2 S \bar{c} (C_{m0} + C_{m\alpha} \alpha + C_{m\delta_E} \delta E) + \frac{1}{4} \rho V_a S \bar{c}^2 (C_{m_q} q) \quad (1.59)$$

$$\tau_{Aero z} = \frac{1}{2} \rho V_a^2 S b (C_{n0} + C_{n\beta} \beta + C_{n\delta_A} \delta A + C_{n\delta_R} \delta R) + \frac{1}{4} \rho V_a S b^2 (C_{n_p} p + C_{n_r} r) \quad (1.60)$$

1.10 Angle and global position of the craft

To relate the velocities to crafts angles and global position requires mapping variables between a body-fixed local frame and an inertial global frame using rotation matrices and kinematic differential equations.

The global velocities used to differentiate the global positions will be obtained rotating the local velocity vector v_1 using the Direction Cosine Matrix denoted as R_{tb}

$$\dot{R}_1 = R_{tb}v_1 \quad (1.61)$$

Similarly a $J(\Theta)$ transformation matrix will be used to map the angular rates to Euler angle derivatives.

$$\dot{R}_2 = J^{-1}(\Theta)v_2 \quad (1.62)$$

This results in component equations for positions:

$$\dot{x}_E = u(\cos\theta\cos\psi) + v(\sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi) + w(\cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi) \quad (1.63)$$

$$\dot{y}_E = u(\cos\theta\sin\psi) + v(\sin\phi\sin\psi\sin\theta + \cos\phi\cos\psi) + w(\cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi) \quad (1.64)$$

$$\dot{z}_E = -u(\sin\theta) + v(\sin\phi\cos\theta) + w(\cos\phi\cos\theta) \quad (1.65)$$

And for rotations:

$$\dot{\phi} = p + q(\sin\phi\tan\theta) + r(\cos\phi\tan\theta) \quad (1.66)$$

$$\dot{\theta} = q(\cos\phi) - r(\sin\phi) \quad (1.67)$$

$$\dot{\psi} = q\left(\frac{\sin\phi}{\cos\theta}\right) + r\left(\frac{\cos\phi}{\cos\theta}\right) \quad (1.68)$$

1.11 Nonlinear state-space model

$$\begin{aligned}
\dot{\xi} = \begin{bmatrix} \dot{x}_E \\ \dot{y}_E \\ \dot{z}_E \\ \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \\ \dot{u} \\ \dot{v} \\ \dot{w} \\ \dot{p} \\ \dot{q} \\ \dot{r} \\ \eta \dot{E}_r \\ \eta \dot{E}_l \end{bmatrix} &= \begin{bmatrix} u(\cos\theta\cos\psi) + v(\sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi) + w(\cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi) \\ u(\cos\theta\sin\psi) + v(\sin\phi\sin\psi\sin\theta + \cos\phi\cos\psi) + w(\cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi) \\ -u(\sin\theta) + v(\sin\phi\cos\theta) + w(\cos\phi\cos\theta) \\ p + q(\sin\phi\tan\theta) + r(\cos\phi\tan\theta) \\ q(\cos\phi) - r(\sin\phi) \\ q(\frac{\sin\phi}{\cos\theta}) + r(\frac{\cos\phi}{\cos\theta}) \\ rv - qw + \frac{X}{m_{craft}} \\ pw - ru + \frac{Y}{m_{craft}} \\ qu - pv + \frac{Z}{m_{craft}} \\ c_1pq + c_2rq + c_7\tau_\phi + c_9\tau_\psi \\ c_3(p^2 - r^2) + c_4pr + c_8\tau_\theta \\ c_5pq + c_6rq + c_9\tau_\phi + c_{10}\tau_\psi \\ clip(\frac{1}{\tau_e}(\delta\eta E_r - \eta E_r), -\omega_{emax}, +\omega_{emax}) \\ clip(\frac{1}{\tau_e}(\delta\eta E_l - \eta E_l), -\omega_{emax}, +\omega_{emax}) \end{bmatrix} \quad (1.69)
\end{aligned}$$

Chapter 2

Trims of the model relevant for transition from horizontal to vertical flight

2.1 Hover trim

The trim at hover is the easiest case to analyze and provides clean closed form solutions. The static equilibrium state of the craft in hover assumes that the craft does not accelerate in any direction and that it stays level with its surroundings:

$$u = v = w = p = q = r = 0$$

$$\phi = \theta = 0$$

In this condition $V_a = 0$ which causes the aerodynamic forces to vanish and aerodynamic moments tend to zero. This reduces finding trim at hover to finding engine command that counteracts the force of gravity.

$$\eta E_r = \eta E_l = 90^\circ$$

With both engines at 90° the forces resulting from engine thrust become:

$$F_{Thrustx} = 0, F_{Thrustz} = -(T_r + T_l)$$

Since the body z -axis points downward, vertical equilibrium for a symmetric solution becomes:

$$T_r = T_l = \frac{mg}{2}$$

Therefor using thrust law as proposed at 1.8 the equilibrium becomes:

$$\delta N_r = \delta N_l = \sqrt{\frac{m_{craft}g}{2C_T\rho D_{prop}^4}}, \eta E_r = \eta E_l = \frac{\pi}{2} \quad (2.1)$$

2.2 Forward, straight level flight trim

The problem of forward straight level flight trim of the proposed model involves determining the steady-state operating condition where all linear and angular accelerations are zero. While for a regular fixed wing craft this becomes a problem of balancing drag with thrust, gravity with lift and momentum with control surfaces the proposed VTOL model creates another two axes of freedom that allow thrust to act in way independent of pitch and angle of attack to counteract both drag and gravity. The considered trim case will be symmetric, meaning that the two additional degrees

of freedom collapse in singular, symmetric engine tilt.

This leaves the system underdetermined as it has fewer independent equations than unknown variables and requires constraining additional parameter beyond V_a . To resolve this the additional degree of freedom given by the tilt rotor design will be prescribed. This will allow a steady-state trim for level, forward trim to be determined and will later allow for finding trims for specific vertical to horizontal flight transition segments. This trim thus imposes following conditions:

$$\begin{aligned} p &= q = r = 0 \\ v &= 0, \beta = 0, \phi = 0 \\ \eta E_r &= \eta E_l = \eta E \\ N_r &= N_l = N \end{aligned}$$

Let the prescribed airspeed be $V_a > 0$. From the definition of angle of attack:

$$\begin{aligned} u &= V_a \cos \alpha \\ w &= V_a \sin \alpha \end{aligned}$$

Level flight requires $\dot{z}_E = 0$. The imposition $\phi = 0$ reduces the kinematics of $\dot{z}_E$ to:

$$\dot{z}_E = -u \sin \theta + w \cos \theta \quad (2.2)$$

Therefore:

$$-V_a \cos \alpha \sin \theta + V_a \sin \alpha \cos \theta = V_a \sin(\alpha - \theta) = 0 \quad (2.3)$$

For this trim thus $\theta = \alpha$. Once α is found the entire kinematic trim state is known:

$$u = V_a \cos \alpha, v = 0, w = V_a \sin \alpha, \phi = 0, \theta = \alpha, p = q = r = 0 \quad (2.4)$$

2.2.1 Longitudinal force equilibrium and angle of attack

Keeping constant airspeed requires the thrust generated by the engines to be in equilibrium with aerodynamic forces opposing them. The symmetrical assumption of the trim allows for the total thrust to be assumed as:

$$P = T_r + T_l = 2T \quad (2.5)$$

The P here is used to denote total thrust from both engines, not power as it would usually be. The force of gravity at $\theta = \alpha$ and $\phi = 0$ becomes:

$$mg = \begin{bmatrix} -\sin \alpha \\ 0 \\ \cos \alpha \end{bmatrix} \quad (2.6)$$

Since $p = q = r = 0$ keeping constant u and w requires the sums of forces X and Z to be both equal to 0 the two longitudinal force equations are:

$$P \cos \eta E - D \cos \alpha + L \sin \alpha - mg \sin \alpha = 0 \quad (2.7)$$

and

$$-P \sin \eta E - D \sin \alpha - L \cos \alpha + mg \cos \alpha = 0 \quad (2.8)$$

These could already be solved as a two-variable nonlinear system in (P, α) , but P can be further eliminated analytically by rotating the force balance into flight-path coordinates. Projecting 2.7-2.8 along the flight path results in:

$$P(\cos\eta E \cos\alpha - \sin\eta E \sin\alpha) - D = 0$$

thus:

$$P \cos(\eta E + \alpha) = D \quad (2.9)$$

Provided that $\cos(\eta E + \alpha) \neq 0$ this becomes:

$$P(\alpha) = \frac{D(\alpha)}{\cos(\eta E + \alpha)} \quad (2.10)$$

Projecting normal to the velocity results in the second equilibrium condition becoming:

$$mg - L - P \sin(\eta E + \alpha) = 0$$

thus:

$$L + P \sin(\eta E + \alpha) = mg \quad (2.11)$$

The P in this equation can be substituted with 2.10 resulting in:

$$L(\alpha) + D(\alpha) \tan(\eta E + \alpha) - mg = 0 \quad (2.12)$$

Let the dynamic pressure and the lift and drag in a level trim be

$$q_\infty = \frac{1}{2} \rho V_a^2, \quad (2.13)$$

$$L(V_a, \theta) = q_\infty S C_l(\alpha)|_{\alpha=\theta}, \quad D(V_a, \theta) = q_\infty S C_d(\alpha)|_{\alpha=\theta}.$$

Following this the aerodynamic forces here can be expanded resulting in final single scalar trim equation:

$$f(\alpha) = q_\infty S (C_l(\alpha) + C_d(\alpha) \tan(\eta E + \alpha)) - mg = 0 \quad (2.14)$$

This equation needs to be solved numerically to determine α and allow the full trim to be computed. An alternative form can also be used to avoid the explicit tangent:

$$\tilde{f}(\alpha) = (L(\alpha) - mg) \cos(\eta E + \alpha) + D(\alpha) \sin(\eta E + \alpha) = 0 \quad (2.15)$$

Once α^* is known P can be calculated as:

$$P^* = \frac{D^*}{\cos(\eta E + \alpha^*)} \quad (2.16)$$

where:

$$D^* = q_\infty S C_d(\alpha)(\alpha^*)$$

Therefor using thrust law as proposed at 1.8 the engine control input becomes:

$$\delta N_r = \delta N_l = \sqrt{\frac{P^*}{2 C_T \rho D_{prop}^4}} \quad (2.17)$$

2.2.2 Pitch moment equilibrium

In order to satisfy the $q = 0$ condition for the proposed model the elevator has to cancel both aerodynamic pitch moment and moment the thrust induced pitch moment. The complete equation for pitch moment equilibrium is:

$$q_\infty S \bar{c} (C_{m_0} + C_{m_\alpha} \alpha^* + C_{m_{\delta_E}} \delta E) + E_{Height} P^* \cos \eta E = 0$$

thus:

$$\delta E = - \frac{C_{m_0} + C_{m_\alpha} \alpha^* + \frac{E_{Height} P^* \cos \eta E}{q_\infty S \bar{c}}}{C_{m_{\delta_E}}} \quad (2.18)$$

Given that the $\dot{\eta} E = 0$ the tilt command becomes:

$$\delta \eta E_r = \delta \eta E_l = \eta E \quad (2.19)$$

2.2.3 Yaw and roll moment equilibrium

For an exactly symmetric state $\beta = p = r = 0$ and equal engine thrust/tilt makes both thrust-induced roll/yaw moments and the differential propeller moments cancel. For a case where the aerodynamic model is perfectly symmetric as well, that is C_{l_0} and C_{n_o} both equal to zero the rudder and aileron commands become:

$$\delta A = \delta R = 0$$

For other cases the trim can still be solved by solving:

$$C_{l_{\delta_A}} \delta A + C_{l_{\delta_R}} \delta R = -C_{l_0}$$

$$C_{n_{\delta_A}} \delta A + C_{n_{\delta_R}} \delta R = -C_{n_o}$$

Thus the control inputs become:

$$\delta A = \frac{-C_{l_0} C_{n_{\delta_R}} + C_{l_{\delta_R}} C_{n_o}}{\Delta} \quad (2.20)$$

$$\delta R = \frac{-C_{l_{\delta_A}} C_{n_o} + C_{n_{\delta_A}} C_{l_0}}{\Delta} \quad (2.21)$$

Where:

$$\Delta = C_{l_{\delta_A}} C_{n_{\delta_R}} - C_{l_{\delta_R}} C_{n_{\delta_A}}$$

2.3 Transition trim manifold

The proposed hover and straight-and-level solutions derived above describe individual equilibrium operating conditions. To study the transition between them, these solutions can be extended into a family parameterized by airspeed and symmetric engine tilt. The objective is to determine which combinations of these two parameters admit a trim, to identify the boundaries imposed by actuator and operating limits, and to determine whether the admissible family contains a continuous connection from hover to forward flight.

More over, if a continuous line connecting hover and forward flight can be created over this manifold then the given craft configuration possesses a continuous transition corridor. This continuous transition corridor implies:

- Viability of gain scheduling - a series of localized Linear Quadratic Regulators can be designed along this line and smoothly blended. The continuous transition corridor guarantees that no sudden jump in plant dynamics will occur that can destabilize the craft.
- Simplified path planning - the high level controller only needs to govern a single virtual progression value from hover to normal flight.
- No "No-Man's Land" - it's guaranteed that through the entire transition the craft can maintain steady-state flight mitigating loss of control.

It should be noted that existence of the continuous transition corridor does not imply stability. An external disturbance can still knock the crafts off its ideal transition trajectory and return to it is not guaranteed.

2.3.1 Definition of the trim family

The two independent parameters are collected in

$$\boldsymbol{\lambda} = (V_a, \eta E), \quad V_a \geq 0. \quad (2.22)$$

At a prescribed parameter pair, the unknown quantities are the pitch attitude, total thrust and elevator deflection:

$$\mathbf{y} = (\theta, P, \delta E), \quad \mathbf{z} = (V_a, \eta E, \theta, P, \delta E). \quad (2.23)$$

Equal propeller speeds and equal engine tilts are assumed. For an exactly symmetric aerodynamic model, zero aileron and rudder deflections satisfy the lateral trim conditions. If the constant roll or yaw moment coefficients are nonzero, the lateral equilibrium equations derived in the preceding trim must also be solved and their actuator limits checked.

The level-flight kinematics can be imposed without using the angle of attack as an independent coordinate:

$$\begin{aligned} u &= V_a \cos \theta, & w &= V_a \sin \theta, \\ v &= \phi = p = q = r = 0, \\ \dot{z}_E &= -u \sin \theta + w \cos \theta = 0. \end{aligned} \quad (2.24)$$

For positive airspeed on the forward-flight branch, these relations imply $\alpha = \theta$ and $\beta = 0$. Pitch attitude remains meaningful at hover, whereas the aerodynamic angles do not. This distinction permits the same configuration variable to be used throughout the construction.

A trim requires constant attitude, body velocity and nacelle tilt. It does not require every inertial position derivative to vanish: a forward-flight trim translates uniformly. The cyclic position coordinates and the arbitrary heading are therefore excluded from the reduced trim coordinates in (2.23).

2.3.2 Implicit equilibrium equations

Let the dynamic pressure and the lift and drag in a level trim be

$$\begin{aligned} q_\infty &= \frac{1}{2} \rho V_a^2, \\ L(V_a, \theta) &= q_\infty S C_l(\alpha)|_{\alpha=\theta}, & D(V_a, \theta) &= q_\infty S C_d(\alpha)|_{\alpha=\theta}. \end{aligned} \quad (2.25)$$

The lift and drag coefficients are evaluated at the angle of attack imposed by the level-flight kinematics, namely $\alpha = \theta$. At zero airspeed, the forces are evaluated through their zero-speed limits.

In level flight, the angle of the total thrust above the horizontal flight path is

$$\chi = \eta E + \theta. \quad (2.26)$$

Projection of the force balance onto the horizontal and vertical directions, together with the pitch-moment balance, gives

$$\mathbf{F}_{\text{trim}}(\mathbf{y}; \boldsymbol{\lambda}) = \begin{bmatrix} P \cos \chi - D \\ L + P \sin \chi - m_{\text{craft}} g \\ q_{\infty} S \bar{c} (C_{m_0} + C_{m_{\alpha}} \theta + C_{m_{\delta_E}} \delta E) + E_{\text{Height}} P \cos \eta E \end{bmatrix} = \mathbf{0}. \quad (2.27)$$

The first row balances horizontal thrust and drag, the second balances upward thrust and lift against weight, and the third imposes zero pitching moment. The symbol E_{Height} must be interpreted as the signed body- z coordinate of the thrust-vector rotation axes relative to the centre of mass. With the body- z axis pointing downward, engines above the centre of mass have $E_{\text{Height}} < 0$.

Equation (2.27) is the primary definition of the trim family. It contains neither a division by dynamic pressure nor a division by the horizontal component of the thrust direction. It is consequently preferable to the explicit forward-flight formulas when approaching hover.

The force equations also admit a useful scalar reduction. Eliminating thrust by taking the component perpendicular to its direction yields

$$H(\theta; V_a, \eta E) = (L - m_{\text{craft}} g) \cos \chi + D \sin \chi = 0. \quad (2.28)$$

For each root, the corresponding thrust is recovered by projection along the thrust direction:

$$P = D \cos \chi + (m_{\text{craft}} g - L) \sin \chi. \quad (2.29)$$

Indeed, the required thrust vector has horizontal and upward components $(D, m_{\text{craft}} g - L)$. Equation (2.28) makes this vector parallel to the prescribed thrust direction, while (2.29) determines its signed magnitude. Roots requiring negative thrust are rejected. This reconstruction remains defined when $\cos \chi = 0$, unlike the expression $D / \cos \chi$.

At positive dynamic pressure and nonzero elevator effectiveness, pitch balance then gives

$$\delta E = - \frac{q_{\infty} S \bar{c} (C_{m_0} + C_{m_{\alpha}} \theta) + E_{\text{Height}} P \cos \eta E}{q_{\infty} S \bar{c} C_{m_{\delta_E}}}. \quad (2.30)$$

This explicit relation is convenient away from hover. Near zero airspeed, the alternative coordinates developed below avoid its vanishing denominator.

2.3.3 Admissibility of an equilibrium

A solution of the equilibrium equations need not be attainable by the actuators. The symmetric propeller speed follows from the assumed thrust law:

$$N = \sqrt{\frac{P}{2C_T \rho D_{\text{prop}}^4}}, \quad \text{RPM} = 60N. \quad (2.31)$$

The conversion factor applies when the numerical value of N is expressed in revolutions per second. A maximum permitted RPM therefore imposes

$$0 \leq P \leq P_{\text{max}}, \quad P_{\text{max}} = 2C_T \rho D_{\text{prop}}^4 \left(\frac{\text{RPM}_{\text{max}}}{60} \right)^2. \quad (2.32)$$

The elevator, nacelle and attitude bounds are imposed separately:

$$\begin{aligned} |\delta E| &\leq \delta_{E,\max}, & \eta_{E,\min} &\leq \eta E \leq \eta_{E,\max}, \\ \theta_{\min} &\leq \theta \leq \theta_{\max}. \end{aligned} \quad (2.33)$$

The elevator bound is assumed symmetric. For unequal positive and negative travel limits, it is replaced by the corresponding interval constraint. Controls must not be clipped after solving the equations, since clipping generally destroys the equilibrium.

For symmetric elevator limits, the existence of an admissible elevator setting is equivalently expressed without division by dynamic pressure:

$$|q_\infty S \bar{c} (C_{m_0} + C_{m_\alpha} \theta) + E_{Height} P \cos \eta E| \leq q_\infty S \bar{c} |C_{m_{\delta_E}}| \delta_{E,\max}. \quad (2.34)$$

The left-hand side is the moment requiring cancellation and the right-hand side is the maximum available elevator moment. Unlike (2.30), this condition remains meaningful at hover.

The selected operating domain must also respect the validity of the aerodynamic and propulsion approximations. In particular, a post-stall lift and drag model does not by itself establish the validity of the linear pitching-moment or elevator-effectiveness coefficients. A conservative admissible domain can therefore restrict angle of attack to the range in which all the coefficients used in the equilibrium are supported. At zero airspeed, no constraint should be interpreted as a measurement of the physically undefined aerodynamic angles.

If the RPM ceiling does not already represent a sustainable loaded operating condition, an available-power limit may also be required. The propeller torque law gives the total shaft-power demand

$$\mathcal{P}_{\text{shaft}} = 2(2\pi N Q_{\text{prop}}) = 4\pi \rho K_q D_{\text{prop}}^5 N^3. \quad (2.35)$$

Here $Q_{\text{prop}} = \rho K_q D_{\text{prop}}^5 N^2$ is the resisting torque of one propeller. Electrical limits must be converted to shaft power using the appropriate efficiency.

2.3.4 Local geometry of the trim manifold

The reduced equilibrium set and its admissible subset are defined by

$$\begin{aligned} \mathcal{M} &= \{\mathbf{z} : \mathbf{F}_{\text{trim}}(\mathbf{z}) = \mathbf{0}\}, \\ \mathcal{M}_{\text{feas}} &= \{\mathbf{z} \in \mathcal{M} : \text{all selected admissibility conditions hold}\}. \end{aligned} \quad (2.36)$$

The notation for the residual denotes the same balances whether its arguments are written as the complete coordinate vector or partitioned into prescribed and dependent variables. If the residual is continuously differentiable and its full Jacobian has rank three, the regular-level-set theorem implies

$$\text{rank } D\mathbf{F}_{\text{trim}} = 3 \implies \dim \mathcal{M} = 5 - 3 = 2. \quad (2.37)$$

Actuator inequalities restrict this surface and may introduce boundaries, corners or disconnected admissible components.

For airspeed and tilt to serve as local coordinates, a stronger condition is needed. Define the dependent-variable Jacobian by

$$\mathbf{J}_y = \frac{\partial \mathbf{F}_{\text{trim}}}{\partial \mathbf{y}} = \frac{\partial \mathbf{F}_{\text{trim}}}{\partial (\theta, P, \delta E)}. \quad (2.38)$$

Where $\det \mathbf{J}_y \neq 0$, the implicit-function theorem gives a locally unique smooth solution $\mathbf{y} = \mathbf{y}(\boldsymbol{\lambda})$. The manifold can then be displayed as a family of trims over the requested airspeed–tilt plane.

This representation need not be globally single-valued. Several roots of (2.28) can exist at one airspeed–tilt pair, with different pitch attitudes, thrust demands or elevator settings. The plot is therefore a projection of the complete equilibrium set, and distinct solution branches must remain distinguishable. A singular dependent-variable Jacobian may indicate a fold of this projection rather than the end of the trim surface.

2.3.5 Regular treatment of hover

The aerodynamic-angle definitions become unsuitable at zero airspeed:

$$\alpha = \text{atan2}(w, u), \quad \beta = \text{atan2}\left(v, \sqrt{u^2 + w^2}\right). \quad (2.39)$$

At hover there is no freestream direction. In addition, near zero speed even small velocity perturbations can produce large angle changes, as illustrated by

$$d\alpha = \frac{u dw - w du}{u^2 + w^2}. \quad (2.40)$$

This expression is valid only where its denominator is nonzero and the angle branch is smooth. The singularity concerns the airflow coordinates, not an infinite physical aerodynamic load.

For bounded force coefficients, the freestream forces vanish as $O(V_a^2)$. The pitch moment can likewise be evaluated in the already rearranged form

$$\begin{aligned} (\tau_{Aero})_y &= \frac{1}{2} \rho V_a^2 S \bar{c} (C_{m_0} + C_{m_\alpha} \alpha + C_{m_{\delta_E}} \delta E) \\ &\quad + \frac{1}{4} \rho V_a S \bar{c}^2 C_{m_q} q. \end{aligned} \quad (2.41)$$

For bounded rates this tends to zero. The implementation should therefore retain Cartesian velocities and attitude, and evaluate the aerodynamic limits directly at zero speed. Replacing airspeed by a positive floor in the dynamic pressure would instead introduce artificial aerodynamic forces and elevator authority.

At zero airspeed, (2.27) reduces to

$$\begin{aligned} P \cos(\eta E + \theta) &= 0, \\ P \sin(\eta E + \theta) &= m_{craft} g, \\ E_{Height} P \cos \eta E &= 0. \end{aligned} \quad (2.42)$$

For nonzero engine height, positive thrust and the upright branch, the solution agrees with the hover trim derived above:

$$\eta E|_H = \frac{\pi}{2}, \quad \theta|_H = 0, \quad P|_H = m_{craft} g, \quad \text{RPM}|_H = 60 \sqrt{\frac{m_{craft} g}{2 C_T \rho D_{prop}^4}}. \quad (2.43)$$

The elevator setting remains undetermined because the assumed unblown elevator generates no moment at hover. Accordingly, the hover boundary is

$$\mathcal{H} = \left\{ \left(0, \frac{\pi}{2}, 0, m_{craft} g, \delta E \right) : |\delta E| \leq \delta_{E, \max} \right\}. \quad (2.44)$$

All of these points project onto the same airspeed–tilt coordinate.

At this boundary, the Jacobian associated with the airspeed–tilt chart is

$$\mathbf{J}_y|_H = \begin{bmatrix} -m_{craft} g & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (2.45)$$

The zero elevator column prevents a locally unique solution for elevator deflection at prescribed airspeed and tilt. To remove this coordinate degeneracy, prescribe airspeed and elevator deflection near hover, and solve for pitch, thrust and tilt instead. The corresponding Jacobian is

$$\left. \frac{\partial \mathbf{F}_{\text{trim}}}{\partial(\theta, P, \eta E)} \right|_{\text{H}} = \begin{bmatrix} -m_{\text{craft}}g & 0 & -m_{\text{craft}}g \\ 0 & 1 & 0 \\ 0 & 0 & -E_{\text{Height}}m_{\text{craft}}g \end{bmatrix}. \quad (2.46)$$

Its determinant is $E_{\text{Height}}(m_{\text{craft}}g)^2$, which is nonzero under the assumed geometry. The full residual therefore retains rank three at hover, even though the airspeed–tilt projection is degenerate. The implicit-function theorem supplies the regular local chart

$$(\theta, P, \eta E) = G(V_a, \delta E). \quad (2.47)$$

This chart parameterizes the same equilibrium surface. Its solutions can still be projected onto the requested airspeed–tilt axes. Actuator inequalities must be checked after continuation; local existence of the unconstrained surface does not ensure local admissibility.

2.3.6 Asymptotic shape of the admissible region near hover

The narrowing of the projected trim region near hover is also a physical consequence of the pitch balance. Define the departure from vertical tilt as

$$\varepsilon = \frac{\pi}{2} - \eta E. \quad (2.48)$$

Along a branch approaching upright hover, total thrust tends to weight and $\cos \eta E = \sin \varepsilon \approx \varepsilon$. The leading-order pitch balance is consequently

$$E_{\text{Height}}m_{\text{craft}}g\varepsilon + \frac{1}{2}\rho V_a^2 S \bar{c} (C_{m_0} + C_{m_{\delta_E}} \delta E) \approx 0. \quad (2.49)$$

With bounded elevator deflection, bounded coefficients and nonzero engine height, this implies

$$\varepsilon = O(V_a^2), \quad V_a \rightarrow 0^+. \quad (2.50)$$

The thrust-induced pitching moment can only grow as quickly as the vanishing elevator authority allows. A uniform tilt grid may therefore fail to resolve the admissible connection near hover even when a continuous branch exists.

For a sufficiently smooth outgoing branch with a finite leading coefficient, write

$$\eta E(V_a) = \frac{\pi}{2} - k_H V_a^2 + o(V_a^2). \quad (2.51)$$

The horizontal and vertical force balances then give

$$\begin{aligned} \theta(V_a) &= \left[k_H - \frac{\rho S}{2m_{\text{craft}}g} C_d(\alpha)|_{\alpha=0} \right] V_a^2 + o(V_a^2), \\ P(V_a) &= m_{\text{craft}}g - \frac{1}{2}\rho S C_l(\alpha)|_{\alpha=0} V_a^2 + o(V_a^2). \end{aligned} \quad (2.52)$$

The pitch balance determines the elevator value approached on that branch:

$$\delta E|_{\text{H}} = - \frac{C_{m_0} + \frac{2E_{\text{Height}}m_{\text{craft}}gk_H}{\rho S \bar{c}}}{C_{m_{\delta_E}}}. \quad (2.53)$$

Thus the allowed branch coefficients satisfy

$$\left| C_{m_0} + \frac{2E_{\text{Height}}m_{\text{craft}}gk_H}{\rho S \bar{c}} \right| \leq |C_{m_{\delta_E}}| \delta_{E,\text{max}}. \quad (2.54)$$

If tilt cannot exceed $\pi/2$, the additional necessary condition is $k_H \geq 0$. Equations (2.51)–(2.53) provide initial predictors for the near-hover calculation. They also show that an arbitrary hover elevator setting need not match the limit of a selected outgoing branch. The elevator may be pre-positioned at hover, where it does not affect the equilibrium in the present model.

2.3.7 Numerical construction of the surface

The surface is constructed using overlapping local charts. The regular airspeed–elevator chart is used first, beginning from the hover boundary and retaining only solutions satisfying the selected limits. Once the dependent-variable Jacobian becomes sufficiently well conditioned, the calculation may switch to the airspeed–tilt chart.

For a regular airspeed–tilt chart, differentiation of the equilibrium gives

$$\mathbf{J}_y d\mathbf{y} + \mathbf{J}_\lambda d\boldsymbol{\lambda} = \mathbf{0}, \quad \mathbf{J}_\lambda = \frac{\partial \mathbf{F}_{\text{trim}}}{\partial \boldsymbol{\lambda}}. \quad (2.55)$$

A first-order prediction at a neighbouring parameter point is therefore

$$\mathbf{y}_{\text{pred}} = \mathbf{y}_k - \mathbf{J}_y^{-1} \mathbf{J}_\lambda \Delta \boldsymbol{\lambda}. \quad (2.56)$$

The Jacobians are evaluated at the current solution, and the prediction is corrected by solving the nonlinear equilibrium equations. In an implementation, the inverse notation represents a linear solve rather than explicit matrix inversion. The same predictor–corrector principle applies in the hover chart after changing the prescribed and dependent variables.

Force and moment residuals have different physical units. A suitable nondimensional residual for convergence checks is

$$\hat{\mathbf{F}}_{\text{trim}} = \begin{bmatrix} (\mathbf{F}_{\text{trim}})_1 / (m_{\text{craft}} g) \\ (\mathbf{F}_{\text{trim}})_2 / (m_{\text{craft}} g) \\ (\mathbf{F}_{\text{trim}})_3 / (m_{\text{craft}} g \bar{c}) \end{bmatrix}. \quad (2.57)$$

Convergence of an optimizer is not sufficient for acceptance: the corrected point must have a sufficiently small equilibrium residual and satisfy all admissibility conditions. Each retained solution should also be substituted into the original dynamics to verify that the required derivatives vanish.

Continuation from one seed does not identify every disconnected branch. At selected parameter pairs, the scalar equation (2.28) should therefore be searched over the accepted pitch interval. Searches must account for tangential roots as well as sign-changing roots. Newly found solutions are retained as separate branches and continued independently. Failure of one local solve is classified as unresolved, not as proof of nonexistence. Resolution is increased near actuator boundaries, folds and closely spaced branches.

At a fold in a fixed-airspeed slice, pseudo-arclength continuation avoids requiring tilt to remain a valid parameter. Collect the continued variables in $\mathbf{w}_c = (\eta E, \theta, P, \delta E)$. In fixed, nondimensionalized continuation coordinates, a unit tangent and a prediction satisfy

$$\mathbf{J}_w \mathbf{t}_c^{(k)} = \mathbf{0}, \quad \|\mathbf{t}_c^{(k)}\| = 1, \quad (2.58)$$

$$\mathbf{w}_c^{\text{pred}} = \mathbf{w}_c^{(k)} + \Delta \ell \mathbf{t}_c^{(k)}.$$

Here the continuation Jacobian differentiates the residual with respect to those scaled coordinates, and the physical variables are recovered before evaluating the force and moment balances. The correction solves

$$\mathbf{F}_{\text{trim}}(\mathbf{w}_c; V_a) = \mathbf{0}, \quad (\mathbf{t}_c^{(k)})^T (\mathbf{w}_c - \mathbf{w}_c^{\text{pred}}) = 0. \quad (2.59)$$

The added condition fixes progress along the branch while allowing tilt to turn back. The same scaling must be used in the tangent, prediction and correction inner product.

2.3.8 A continuous path from hover to forward flight

After the admissible surface has been constructed, a transition schedule is sought as a curve in the complete trim coordinates:

$$\Gamma(s) = (V_a(s), \eta E(s), \theta(s), P(s), \delta E(s)), \quad s \in [0, 1]. \quad (2.60)$$

The pointwise and endpoint conditions are

$$\begin{aligned} \Gamma(s) &\in \mathcal{M}_{\text{feas}}, & \Gamma(0) &\in \mathcal{H}, & \Gamma(1) &\in \mathcal{C}, \\ \mathcal{C} &= \{\mathbf{z} \in \mathcal{M}_{\text{feas}} : \eta E = 0, V_a \in I_C\}. \end{aligned} \quad (2.61)$$

The cruise interval may be replaced by one prescribed final airspeed. Zero engine tilt does not imply zero pitch attitude: the forward-flight endpoint must retain the angle of attack required by its force balance.

The existence of a projected route in the airspeed–tilt plane is insufficient. The route must lift to a continuous curve on an admissible branch in the full trim coordinates. Two projected branches may intersect while requiring different thrust, pitch or elevator settings, so they cannot be joined merely because their plotted coordinates coincide.

A practical initial path is obtained from a graph whose vertices store complete trim solutions. Edges are admitted only when local continuation establishes an admissible connection; the intermediate states must be checked, not merely the endpoints. Graph search then identifies a route from the hover boundary to the selected cruise set. A useful edge cost is

$$c_{ij} = w_\ell \|\bar{\mathbf{z}}_j - \bar{\mathbf{z}}_i\| + w_u \|\bar{\mathbf{a}}_j - \bar{\mathbf{a}}_i\|^2 + w_r \Pi_{ij}. \quad (2.62)$$

Here $\mathbf{a} = (N, \delta E, \eta E)$ contains physical trim settings, overbars indicate fixed nondimensionalization, and the final term penalizes low reserve along the connection. For example, the reserve measures may be defined as

$$\mu_T = \frac{P_{\max} - P}{P_{\max}}, \quad \mu_E = 1 - \frac{|\delta E|}{\delta E_{\max}}, \quad \mu = \min(\mu_T, \mu_E). \quad (2.63)$$

These quantities measure proximity to selected actuator limits, not complete control authority.

The graph route can be refined into a smooth curve by minimizing

$$\min_{\Gamma} \int_0^1 \left[w_1 \|\bar{\Gamma}'(s)\|^2 + w_2 \|\bar{\Gamma}''(s)\|^2 + \frac{w_3}{[\epsilon_J + \mu(\Gamma(s))]^2} \right] ds, \quad (2.64)$$

subject to (2.61) and the equilibrium equations at every point. The positive constant ϵ_J regularizes the penalty. The three terms penalize rapid variation, curvature and low actuator reserve. If monotonically increasing airspeed is required, impose

$$V_a'(s) \geq 0. \quad (2.65)$$

Monotone tilt may also be imposed, but it is an additional restriction and is not guaranteed by connectedness of the admissible surface.

Interpolation should be performed in the independent coordinates of each local chart, with the dependent variables recovered from the equilibrium equations. An arbitrary spline through all five coordinates generally leaves the surface. The airspeed–elevator chart provides the regular interpolation near hover, while the airspeed–tilt chart is used where it is nonsingular.

2.3.9 Operational interpretation of the trim path

For an operational schedule, the equilibrium and travel-limit checks must be supplemented by a control-authority assessment. Linearizing the relevant reduced dynamics, including nacelle dynamics, gives

$$\Delta \dot{\mathbf{x}}_r = \mathbf{A}(\mathbf{z})\Delta \mathbf{x}_r + \mathbf{B}(\mathbf{z})\Delta \mathbf{a}. \quad (2.66)$$

The selected controller must stabilize the required variables with the available actuator margins. An open-loop unstable trim is not automatically excluded, but solvability of the trim equations does not establish stabilizability.

At hover the thrust-induced pitch moment has sensitivity

$$(\tau_{Thrust})_y = E_{Height} P \cos \eta E, \quad \left. \frac{\partial (\tau_{Thrust})_y}{\partial \eta E} \right|_H = -E_{Height} m_{craft} g. \quad (2.67)$$

If both nacelles are restricted to the interval from zero to $\pi/2$, their combined thrust moment has only one sign at zero airspeed, while the unblown elevator contributes no moment. Such a hover equilibrium must not be treated as having two-sided pitch authority. Tilt travel beyond vertical or another effective pitch actuator is needed for that authority in the assumed geometry.

Finally, a continuous trim curve is a sequence of equilibria rather than an accelerating trajectory. During a level transition, the horizontal balance is

$$P \cos(\eta E + \theta) - D = m_{craft} \dot{V}_a. \quad (2.68)$$

Consequently, changing airspeed requires departure from the exact trim state-input pair. Choosing a time law $s = s(t)$ also requires actuator-rate checks, including

$$|\eta E'(s) \dot{s}| \leq \omega_{emax}. \quad (2.69)$$

For the unsaturated first-order nacelle model, the corresponding feedforward tilt command is

$$\delta \eta_{E,cmd} = \eta E(s) + \tau_e \eta E'(s) \dot{s}, \quad (2.70)$$

with command travel limits enforced as well.

The trim curve can initialize a dynamically constrained optimization such as

$$\min_{\xi(t), \mathbf{a}(t), s(t), T_f} \int_0^{T_f} [\|\xi(t) - \xi_{trim}(s(t))\|_{\mathbf{Q}}^2 + \|\mathbf{a}(t) - \mathbf{a}_{trim}(s(t))\|_{\mathbf{R}}^2] dt, \quad (2.71)$$

subject to $\dot{\xi} = \mathbf{f}(\xi, \mathbf{a})$, the actuator and operating constraints, and hover and cruise endpoint conditions. In this dynamic problem the actuator vector contains commands; the weighted norm is $\|b\|_W^2 = b^T W b$. Position references must be supplied kinematically or assigned zero tracking weight, since cyclic positions are not specified by the trim manifold. The transition duration should be fixed or appropriately bounded; the tracking objective alone need not select a finite optimum. Cartesian velocity states are retained through hover.

If the hover boundary and the selected cruise set lie in different path-connected components of the admissible trim set, no continuous steady-trim schedule joins them under the selected constraints. This does not by itself exclude a dynamically feasible transition that departs from those equilibria. Failure of a numerical path search alone does not establish such a separation.